

Shopping Behavior Analysis

Python (EDA) · SQL · Power BI · Business Recommendations



Project Overview

3,900

Transactions

Individual purchase records in the raw dataset

18»→20

Columns

Raw attributes expanded with engineered features

4

Categories

Clothing, Footwear, Accessories, Outerwear

\$233K

Total Revenue

Across all 3,900 transactions

3,900 Transactions	18 → 20 Raw → Engineered Columns	4 Product Categories
25 Unique Items	50 Customer Locations	\$233,081 Total Revenue

An end-to-end retail analytics study.

This project delivers a complete analysis combining Python-based EDA, SQL business queries, an interactive Power BI dashboard, and data-driven recommendations — to understand who customers are, what they buy, and where the business should focus.

Dataset & Cleaning

Dataset: customer_shopping_behavior.csv

- 3,900 transactions · 18 raw attributes
- Covers demographics, product details, pricing, shipping, discounts, ratings & purchase history
- 37 nulls in Review Rating imputed using per-category median
- Confirmed **Discount Applied = Promo Code Used** for every row; dropped redundant column

Feature Engineering (Python)

Column names standardized to snake_case. Two features engineered: **age_group** via quartile binning (pd.qcut) into Young Adult, Adult, Middle-aged, Senior; and **purchase_frequency_days** mapping stated frequency labels to numeric day counts. Cleaned data exported to PostgreSQL via SQLAlchemy.



Exploratory Data Analysis

Key distributions and relationships visualized with pandas and matplotlib across 3,900 transactions.

- Revenue by Gender

Male customers generate **\$157,890 (68%)** of total revenue vs. \$75,191 (32%) for female — approximately 2× more.

- Revenue by Category

Clothing dominates at **~\$105K**, followed by Accessories (~\$75K), Footwear (~\$35K), and Outerwear (~\$18K).

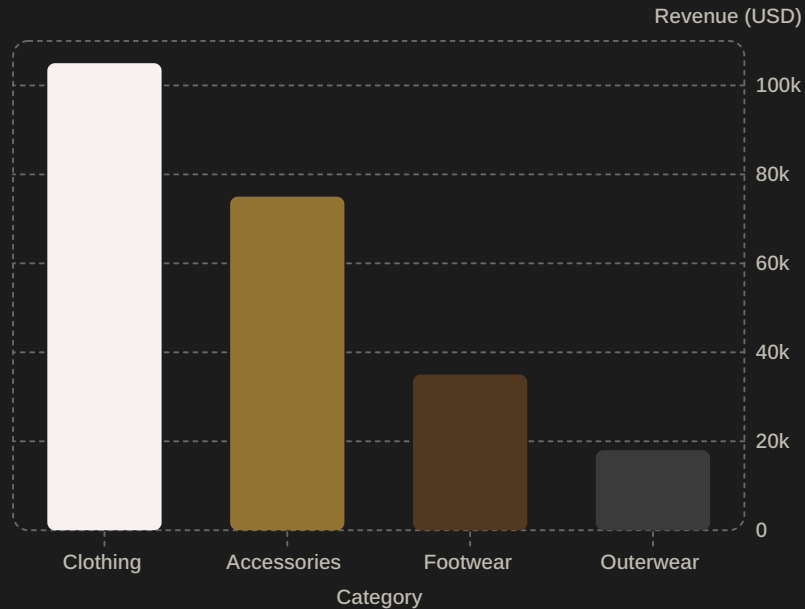
- Spend Distribution

Average purchase is **\$59.76** (median \$60) — a tight, symmetric distribution. Spend is stable (~\$59–60) across all four age groups and shows essentially no correlation with age, rating, or purchase history.

- Subscription Penetration

Only **27%** of customers are active subscribers, yet non-subscribers already generate the majority of revenue — subscription is under-penetrated relative to its potential value.

Business Questions: Revenue & Behavior



Clothing is the dominant revenue category. Outerwear has the most growth headroom — a bundling opportunity at checkout.

01

Q1 – Revenue by Gender

Male: \$157,890 · Female: \$75,191. Male customers account for 68% of total revenue.

02

Q2 – Discount Buyers Above Average

839 customers used a discount and still spent at or above \$59.76 average — discounting is not simply attracting price-sensitive buyers.

03

Q4 – Shipping Impact on Spend

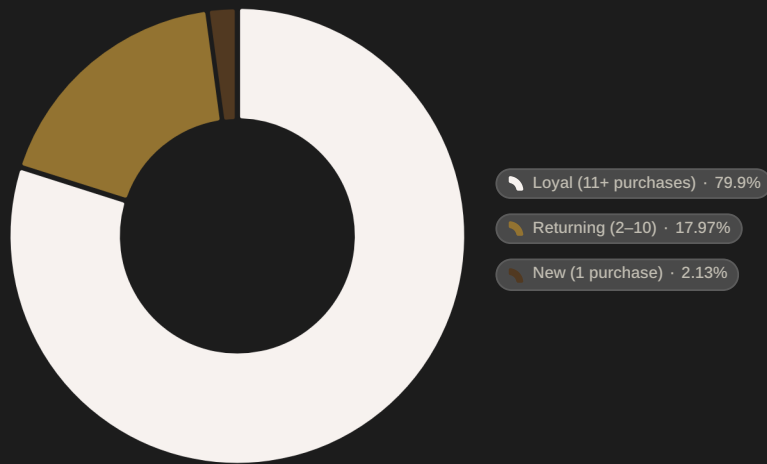
Express shipping customers average \$60.48 vs. \$58.46 for Standard — a \$2 gap per order that can be leveraged as an upsell.

04

Q5 – Subscribers vs. Non-Subscribers

Subscribers (1,053) and non-subscribers (2,847) spend virtually the same per transaction (~\$59.49 vs. \$59.87). Subscription drives frequency, not order value.

Customer Segmentation & Product Insights



An overwhelming **80% of the customer base qualifies as Loyal** — retention economics matter far more here than new-customer acquisition.

01

Q6 – Most-Discounted Items

Hat (50%), Sneakers (49.7%), Coat (49.1%), Sweater (48.2%), and Pants (47.4%) — nearly half of all purchases discounted on these items.

02

Q8 – Top Products per Category

Blouse & Pants lead Clothing (171 orders each); Jewelry leads Accessories (171); Sandals lead Footwear (160); Jacket leads Outerwear (163).

03

Q9 – Repeat Buyers & Subscriptions

2,518 repeat buyers (5+ purchases) are still non-subscribers — high purchase frequency does not auto-convert to subscription sign-up. Clear conversion opportunity.

04

Q10 – Revenue by Age Group

Young Adult \$62,143 · Middle-aged \$59,197 · Adult \$55,978 · Senior \$55,763. Revenue is remarkably flat — the base is broad, not age-concentrated.

Power BI Dashboard

A single interactive page titled "**Customer Behavior Dashboard**" — connect dashboard.pbix to the public.customer PostgreSQL table to explore it live.

1

KPI Cards

Avg Purchase Amount, Avg Review Rating, and Customer Count — update live as slicers are applied

2

Paired Visuals

Revenue vs. customer count shown side-by-side for Category and Age Group — spots high-traffic but low-spend segments instantly

3

4 Cross-Filters

Slicers for Subscription Status, Gender, Category, and Shipping Type — filter the entire page simultaneously





Business Recommendations

Grow Subscription Deliberately

Target the 2,518 non-subscribed repeat buyers first. Bundle with a tangible per-order benefit (early access, free express shipping) rather than a passive badge.

Protect Margin on Discounts

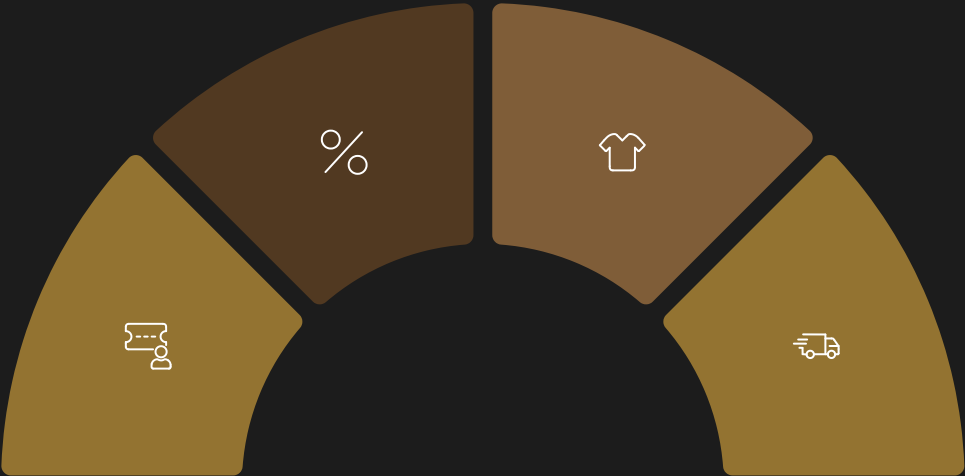
Hat, Sneakers, Coat, Sweater, Pants are discounted on ~50% of purchases. Shift from blanket discounting to loyalty-tier pricing — 839 discount users already spend above average.

Double Down on Clothing

Clothing drives the most revenue. Bundle Outerwear (smallest category) with top Clothing items at checkout to lift average order value.

Upsell Express Shipping

Express customers spend \$2 more per order than Standard. Test express as a default upsell or threshold-based free perk at checkout.



More Recommendations

Three additional strategic actions to guide ongoing decisions.



Retention-First, Not Acquisition-First

80% of customers are already Loyal (11+ purchases). Prioritize service quality, loyalty rewards, and personalized offers over new-customer discounting.



Use Ratings to Guide Merchandising

Top-rated items — Gloves (3.86), Sandals (3.84), Boots (3.82) — are accessory/footwear items. Feature them prominently in marketing to reinforce trust and monitor lower-rated high-volume lines for quality issues.



Target by Behavior, Not Age

Revenue and average spend are nearly flat across all four age groups. Allocate marketing spend by subscription status, purchase frequency, and category affinity — not age bracket.



Summary & Project Files

This analysis reveals a **broad, stable customer base** (\$233K in total revenue across 3,900 transactions) whose spending is remarkably consistent across age groups and only weakly related to review satisfaction or purchase history.

The clearest opportunities are **structural rather than demographic**: convert loyal non-subscribed repeat buyers into subscribers; rationalize heavy discounting on a handful of items; and use shipping-tier and category bundling to lift average order value.

Python EDA

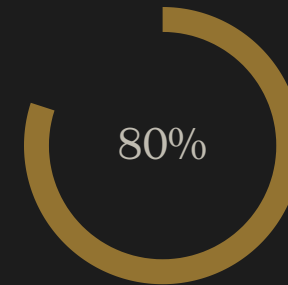
pandas_file.ipynb — cleaning, feature engineering & PostgreSQL export

SQL Queries

customer_database.sql — 10 core business questions answered

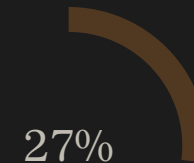
Power BI

dashboard.pbix — interactive KPI and self-service filtering dashboard



Loyal Customers

Already 11+ previous purchases — retention is the primary growth lever



Subscription Rate

Under-penetrated — significant conversion opportunity among repeat buyers



Discount Rate

On top items like Hat and Sneakers — margin at risk without targeted strategy